

Saronik Pal

Kolkata, MI, India

saronik_pal2004@gmail.com

Summary

I am an Electronics Engineer focused on quantum chip development, nanoelectronics, and advanced system design. I am passionate about translating quantum physics into scalable hardware through nanofabrication and precision engineering. I work at the intersection of semiconductor technology, device characterization, and quantum system integration. I aim to develop full-stack quantum-classical frameworks that enable efficient and fault-tolerant integration of logical quantum algorithms with physical hardware

Experience & Projects

Compilers for trapped-ion quantum systems

Self Research, Remote Jan 2026
Investigated Quantum Charge-Coupled Device (QCCD)-based compilers for trapped-ion quantum systems, compared quantum compilers for Hamiltonian simulation circuits, and explored systematic initial quantum state preparation for QAOA.

Analysis of a Superconducting Quantum Chip Structure

Self Research Feb 2026
Study of a quantum chip architecture using HFSS & Qiskit Metal

Design of Quantum Chip

Qiskit Metal (IBM), Remote Aug 2025 — Sep 2025
Multi-Qubit Quantum Processor -
Design a Qubit chip with coupling resonators and shared control infrastructure.

Quantum noise in time-dependent nanoelectronic devices

Self Research May 2025 — Aug 2025
Disorder and Localization in Nanostructures -
By Understanding decoherence and noise in quantum hardware using TKWANT

Quantum Approach to Delaunay Triangulation

Aqora Aug 2025 — Sep 2025
Optimizing complex combinatorial problems such as Delaunay triangulation using **quantum annealing** using D-Wave machines

Multidimensional Regression on LHC collision jets

CERN(Aqora) Jun 2025 — Jul 2025
Modelling the behavior of particle jets from LHC collisions using advanced Quantum Algorithms to bridge theoretical predictions with observed data.

Simulation Engineer

Dassault Systemes - Catia, Remote Oct 2024 — Nov 2024
Using CATIA to simulate stress, strain, and thermal behavior of fabricated parts under real-world conditions, ensuring structural integrity and compliance with manufacturing standards.

Qiskit Advocate

IBM, Remote Jan 2026

- Qiskit Metal
- KWANT & TKWANT(Quantum transport simulations)
- Quantum Algorithms
- Simulation and Modeling
- Quantum System Design (QBraid)
- Multiphysics Integration (Ansys HFSS)

Achievements

1. Top 200 in Physics Brawl (FYKOS) 2024
2. Silver Honour in IAAC 2025
3. Eigen Trails (Qiskit Fall Fest) - IISC - Top 5
4. Diploma in MIT IQHack Hackathon
5. Diploma in Physics Cup (Taltech) – 2024 [Finished Among Top 10]

Websites and Social Links

GitHub:

<https://github.com/PalSaronik01>